

MATH 6112. FALL 2026. ASSIGNMENT 1.

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Problem 1

Exercise 1.3.13-1

Suppose there are two different minimal polynomials p_1, p_2 of T . We use $\deg f$ to denote the degree of polynomial f . By minimality, $\deg p_1 = \deg p_2$. Since both of p_1 and p_2 are monic, we have that $\deg(p_1 - p_2) < \deg p_1$ and $(p_1 - p_2)(T) = p_1(T) - p_2(T) = 0$. Since $p_1 - p_2 \neq 0$, it contradicts to the minimality of p_1 . Therefore, the minimal polynomial is unique.

Exercise 1.3.13-2

Let $p = a_0 + a_1x + \cdots + a_kx^k$ be the minimal polynomial ($a_k = 1$). It is trivial that $k \geq 1$.

$\implies$: Suppose $a_0 = 0$, then we have that

$$0 = T^{-1}p(T) = \sum_{i=1}^k a_i T^{i-1} =: q(T)$$

where $q(x) = a_1 + a_2x + \cdots + a_kx^{k-1}$. $q \neq 0$ since $k \geq 1$. q has lower degree of p and $q(T) = 0$, this contradicts to the minimality of p . Hence, $a_0 \neq 0$.

$\Leftarrow$: Since $p(T) = 0$, we have that

$$T \sum_{i=1}^k \frac{a_i}{a_0} T^{k-i} = I.$$

We complete the proof by setting $T^{-1} = \sum_{i=1}^k \frac{a_i}{a_0} T^{k-i}$.

Exercise 1.3.13-6

(a): $Tv = \alpha v$ implies that $0 = T^k v = \alpha T^{k-1} v = \cdots = \alpha^k v$. Since $v \neq 0$, $\alpha^k = 0$ shows that $\alpha = 0$.

(b): WLOG, we may assume $p = k - 1$ ($T^k = 0$ and we can set $\alpha_i = 0$ for $i > 0$). Construct $\{\beta_i\}_{i \in [k-1] \cup \{0\}}$ as follows,

$$\beta_0 = \frac{1}{\alpha_0} \quad \text{and} \quad \forall i \in [k-1], \beta_i = -\frac{1}{\alpha_0} \sum_{j=1}^i \alpha_j \beta_{i-j}. \quad (1)$$

Set $P = \sum_{i=0}^{k-1} \beta_i T^i$. By Equation (1), $SP = I + T^k H = I$, where H is some polynomial of T . Therefore, S is invertible and $S^{-1} = P$.

Problem 2

We first prove that C is invertible. Let $V = (v_1, \dots, v_n)$ and $W = (w_1, \dots, w_n)$. Let D be the matrix associated to the linear transformation which changes the basis: $v_j = \sum_{i=1}^n c_{ij} w_i$. We have that $VC = W$ and $WD = V$. Therefore, $VCD = WD = V$. Since $v_1 \dots v_n$ are linearly independent, $CD = I$ and C is invertible.

We then prove that $B = DAC$. The following holds:

$$\begin{aligned}
 WDAC &= VAC = [T(v_1), \dots, T(v_n)]C && (\text{def. of } A) \\
 &= \left[\sum_{i=1}^n c_{1i} T(v_i), \sum_{i=1}^n c_{2i} T(v_i), \dots, \sum_{i=1}^n c_{ni} T(v_i) \right] \\
 &= \left[T\left(\sum_{i=1}^n c_{1i} v_i\right), T\left(\sum_{i=1}^n c_{2i} v_i\right), \dots, T\left(\sum_{i=1}^n c_{ni} v_i\right) \right] && (T \in \text{Hom}(V, V)) \\
 &= [T(w_1), \dots, T(w_n)] = WB. && (\text{def. of } B)
 \end{aligned}$$

Since $w_1, \dots, w_n$ are linearly independent, we have that $DAC = B$.

Problem 3

(a): By Block Diagonal Form (Corollary 1.5.45), there exists V such that $V^{-1}AV = R = \text{diag}\{R_{ii}\}_{i \in [p]}$, where $R_{ii} = \lambda_i I + U_{ii}$ and $U_{ii} \in \mathbb{C}^{m_i \times m_i}$ is strictly upper triangle. Fix $k \in [p]$, let $U_{kk} = (u_{ij})_{i,j \in [m_k]}$ and $v_1 \dots v_{m_k}$ be the $(1 + \sum_{i < k} m_i)$ -th to $(\sum_{i \leq k} m_i)$ -th column vectors of V . Since $AV = VR$, we have that

$$\begin{cases} Av_1 = \lambda_k v_1 \\ Av_2 = u_{12}v_1 + \lambda_k v_2 \\ \vdots \\ Av_{m_k} = \lambda_k v_{m_k} + \sum_{i < m_k} u_{im_k} v_i \end{cases} \implies \begin{cases} (A - \lambda_k I)v_1 = 0 \\ (A - \lambda_k I)^2 v_2 = u_{12}(A - \lambda_k I)v_1 = 0 \\ \vdots \\ (A - \lambda_k I)^{m_k} v_{m_k} = (A - \lambda_k I)^{m_k-1} \sum_{i < m_k} u_{im_k} v_i = 0 \end{cases}$$

Therefore, $v_1, \dots, v_{m_k}$ are solutions of $(A - \lambda_k I)^{m_k} v = 0$ and the solution set is spanned by at least m_k linearly independent vectors. On the other hand, let v be one column vector of V and $v \notin \{v_1, \dots, v_{m_k}\}$. Suppose λ be the corresponding eigenvalue of v , we have that

$$(A - \lambda_k I)^{m_k} V = V(R - \lambda_k I)^{m_k} \implies$$

$$(A - \lambda_k I)^{m_k} v = (\lambda - \lambda_k)^{m_k} v + \text{linear combination of column vectors of } V \neq 0,$$

since $\lambda - \lambda_k \neq 0$ and linear independence of column vectors of V . Since $\dim(V) = n$, the dimension of solution set of $(A - \lambda_k I)^{m_k} v = 0$ is m_k .

(b) Putting together the sets of vectors from (a), we get exactly V . Since $\dim(V) = n$, it is a basis of $\mathbb{C}^n$.

Problem 4

Problem 5

We prove by induction on k . The statement follows from definition of Jordan block if $k = 1$.

Now suppose $k > 1$. We have that

$$\begin{aligned}
(J^k)_{ij} &= \sum_{\ell=1}^n (J^{k-1})_{i\ell} (J)_{\ell j} = \sum_{\ell=i}^{i+k-1} \binom{k-1}{\ell-i} \lambda^{k-1-(\ell-i)} (J)_{\ell j} & (\text{I.H.}) \\
&= \lambda^{k+i-j} \sum_{\ell=i}^{i+k-1} \mathbb{1}[j-1 \leq \ell \leq j] \binom{k-1}{\ell-i} & (\text{def. of } J) \\
&= \begin{cases} \binom{k-1}{j-i-1} + \binom{k-1}{j-i} & , i \leq j-1 \wedge j \leq i+k-1 \\ \binom{k-1}{j-i} = 1 = \binom{k}{j-i} & , j = i \\ \binom{k-1}{j-i-1} = 1 = \binom{k}{j-i} & , j = i+k \\ 0 & , \text{o.w.} \end{cases} & (2)
\end{aligned}$$

Since $\binom{k-1}{j-i-1} + \binom{k-1}{j-i} = \binom{k}{j-i}$, [Equation \(2\)](#) completes the proof.

Problem 6

By definition we have that

$$\begin{aligned}
\frac{d(\det A(\mu))}{d\mu} &= \sum_{\sigma \in S} (-1)^\sigma \frac{d}{d\mu} \prod_{i=1}^n a_{i,\sigma_i} = \sum_{\sigma \in S} (-1)^\sigma \sum_{j=1}^n a'_{j,\sigma_j} \prod_{i \in [n], i \neq j}^n a_{i,\sigma_i} \\
&= \sum_{j=1}^n \sum_{\sigma \in S} (-1)^\sigma a'_{j,\sigma_j} \prod_{i \in [n], i \neq j}^n a_{i,\sigma_i} = \sum_{j=1}^n \det(a_1, \dots, a_{j-1}, a'_j, a_{j+1}, \dots, a_n).
\end{aligned}$$

Problem 7